Embedded Protection System – Progress Report

Real-time 3-Phase Pump Power Abnormality Monitoring & Protection

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1. Project Overview

This project develops a low-voltage embedded monitoring and protection system for Jahan Pump TYPE 4 series three-phase submersible drainage pumps (typical operating range 10–30 A). The system measures voltage and current on all three phases, detects dangerous conditions (over-current, over/under voltage, phase loss), and activates a relay that will later interrupt the contactor coil circuit. A manual reset is required after every trip. The work is intended as a professional portfolio project for an Embedded Systems Master's application.

2. Hardware Platform

Category	Component	Details
Controller	ESP32-WROOM-32	Dual-core, Wi-Fi, I2C, ADC
Precision ADC	ADS1115	16-bit, used for all three current channels
Current Sensors	3 × SCT-013 100A/50mA	33 Ω burden + bias circuit → ADS1115
Voltage Sensors	3 × ZMPT101B	Connected to ESP32 ADC pins 35, 32, 33
Actuation	5 V Relay Module	TONGLING JQC-3FF (opto-isolated)
User Interface	OLED + LEDs + Button	0.96" SSD1306, Red/Yellow/Green LEDs, Reset button
Power	ATX 5 V + 1 A fuse	Fused and decoupled low-voltage supply

3. Assembly & Firmware Progress

Stage	Description	Status
1	Power supply (ATX 5 V + fuse)	Completed
2	Main controller (ESP32 + OLED + LEDs + Button + Relay)	Completed
3	Current sensors (3 × CT + burden + bias)	Completed
4	Voltage sensors (3 × ZMPT101B)	Completed
5	ADS1115 integration + 6-channel routing	Completed
6	Basic protection logic + startup delay + trip/reset	Completed
7	Code structured for calibration	Completed
8	Real-pump calibration & contactor interface	Next

4. Final Sensor Assignment

Signal	Sensor	Connected To
Current Phase 1	SCT-013 + bias	ADS1115 AIN0
Current Phase 2	SCT-013 + bias	ADS1115 AIN1
Current Phase 3	SCT-013 + bias	ADS1115 AIN2
Voltage Phase 1	ZMPT101B	ESP32 GPIO 35
Voltage Phase 2	ZMPT101B	ESP32 GPIO 32
Voltage Phase 3	ZMPT101B	ESP32 GPIO 33

5. Current Firmware Features

- Live reading of all 6 channels (3 currents + 3 voltages)
- OLED status display (OK / STARTING / TRIPPED)
- Over-current protection (limit currently set to 32 A)
- Over/under voltage protection (180–250 V window)
- 4-second startup delay to ignore motor inrush current
- Trip state with red LED + relay activation
- Manual reset via push button
- Calibration factors placed at the top of the code for easy adjustment

6. Next Steps

- Calibrate current and voltage readings using a real TYPE 4 pump
- Safely insert the relay into the contactor coil circuit
- Perform end-to-end protection tests on a live pump
- Final code cleanup, documentation and GitHub repository

Report updated 3 October 2026. Covers only the embedded low-voltage protection system. Power and control circuit panel assembly is handled separately.